

Sahil

MACHINE LEARNING RESEARCHER | VISION-LANGUAGE MODELS | REINFORCEMENT LEARNING
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Research Summary

Machine learning researcher and team lead with over four years of experience developing large-scale Vision-Language Models and Large Language Models for medical imaging. Currently leading a team of five at Qure AI to build CT and X-ray VLMs for radiology auto-reporting and disease-specific detection across Pulmonary Hypertension, COPD, Lung Cancer, and Heart Failure. Experience spans data curation, pretraining, supervised fine-tuning, reinforcement-learning-based post-training, clinical validation, and deployment on datasets of more than 10 million medical images. Broader research background includes GANs, super-resolution, inverse reinforcement learning, robust graph neural networks, and uncertainty estimation in medical image segmentation.

Research Interests

Large Language Models, Vision-Language Models, multimodal reasoning, reinforcement learning and model post-training, generative modeling, medical imaging, radiology AI, and clinical model validation.

Education

Indian Institute of Science Education and Research

BS-MS, Electrical Engineering and Computer Science

Bhopal, India

July 2017 – June 2022

Research Experience

Qure AI

Associate Director, Data Science

Bengaluru, India

May 2022 – Present

- **Team leadership:** Lead a team of five and own the full lifecycle of medical Vision-Language Models for CT and X-ray imaging, from data curation and model training through post-training, clinical validation, and deployment.
- **Large-scale VLM development:** Built CT and X-ray VLMs for full radiology auto-reporting and disease-specific detection across Pulmonary Hypertension, COPD, Lung Cancer, and Heart Failure.
- **Pretraining and fine-tuning:** Trained medical VLMs on more than 10 million X-rays using Qwen2.5-VL, InternVL3, Florence-VL, and LLaVA backbones with MS-Swift and custom multimodal data pipelines.
- **Reinforcement-learning post-training:** Designed and ran offline and online reinforcement-learning pipelines using DPO, GRPO, MPO, and GSPO to improve clinical reasoning, disease identification, and structured report generation.
- **Distributed training systems:** Built scalable post-training pipelines using verl and NVIDIA NeMo-RL, achieving state-of-the-art results on full radiology auto-reporting tasks.
- **Clinical deployment:** Developed and deployed the Pulmonary Hypertension VLM at Merck sites and the COPD VLM at CIPLA sites, with multicenter validation against spirometry and specialist review.
- **Product impact:** Developed qMSK, a fracture detection and classification system deployed at Massachusetts General Hospital and Erasmus University; published in ACR, with CE and FDA approvals underway.

Intel Corporation

Graduate Technical Intern

Bengaluru, India

May 2021 – May 2022

- **Applied machine learning:** Developed and deployed machine learning and deep learning models in the Client Computing Group, spanning computer vision, natural language processing, and broader ML applications.

Robert Bosch Engineering and Business Solutions

Research Intern

Bengaluru, India

Jan 2021 – May 2021

- **Robotic computer vision:** Generated synthetic images from robotic-vision datasets using GAN-based methods, including FineGAN and MixAndMatch.

Jio Artificial Intelligence and Center of Excellence

Computer Vision Summer Intern

Hyderabad, India

May 2020 – Aug 2020

- **Generative vision models:** Trained GAN models for face frontalization, generated multi-view images from a single input using HoloGAN, and produced high-resolution barcode images using deep unfolding networks for super-resolution.

Academic Research Experience

University of Alberta

Research Intern

Remote

Aug 2021 – May 2022

- **Inverse reinforcement learning:** Developed inverse reinforcement learning methods from human feedback, targeting faster convergence and improved explainability.

Technical University of Darmstadt

Long-Term Project Student, MEC Lab

Germany

Dec 2020 – June 2021

- **Robust GNNs and medical segmentation:** Developed robust Graph Neural Networks using Capsule Networks and Bayesian methods; studied uncertainty estimation in medical image segmentation using PhiSeg, Stochastic Segmentation Networks, and the Probabilistic U-Net.

Simula Research Laboratory

Long-Term Project Student

Norway

Mar 2020 – Sep 2020

- **GANs and adversarial training:** Trained an ACGAN on CIFAR-10 for synthetic image generation and ResNet classification; implemented BigGAN, StyleGAN, and U-Net GAN variants for synthetic image generation and classification on Caltech-101.

Indian Institute of Science

Computer Vision Summer Intern

Bengaluru, India

- **Monocular depth estimation:** Worked on Struct2Depth and unsupervised monocular ego-motion learning with structure and semantics under Dr. Ragu Krishnapuram, with modifications based on DeepCaps.

Indian Statistical Institute

Summer Research Intern

Kolkata, India

- **Clustering and generative models:** Implemented k-means, kernel k-means, DC-GAN, and W-GAN on MNIST; explored SRGAN modifications using VQ-VAE, VQ-VAE-2, and AE-GAN for high-resolution image generation.

Publications and Abstracts

Peer-Reviewed Papers

- [J1] “AI-based fracture classification and localization on radiographs: cross-validation in a Dutch population.”
Insights into Imaging.
- [C1] “FoCA: Failure-oriented Class Augmentation for Robust Image Classification.”
ICTAI 2022.

Selected Abstracts

- [A1] “Opportunistic COPD detection from routine chest imaging using a Vision-Language AI model.”
ERS 2026, accepted.
- [A2] “Automated multi-fracture detection from X-rays using a multi-task deep neural network.”
RSNA 2023.
- [A3] “Fracture classification and segmentation using a semi-supervised multi-task ensemble.”
EUSOMI 2023.
- [A4] “Deep learning-based fracture detection from musculoskeletal X-ray images.”
ACR 2023.
- [A5] “Deep learning for musculoskeletal abnormality classification in radiographic imaging.”
ECR.

Patents

- Three granted US patents in musculoskeletal imaging.
- Patents in preparation covering VLM-based Heart Failure, COPD, Pulmonary Hypertension, and radiology auto-reporting.

Honors and Awards

- Qualified JEE-Advanced, 2017.
- DST-INSPIRE Scholarship for the study of basic sciences.
- Letter of appreciation from the HRD Minister for performance in the 10th CBSE examination.

References

Prof. Swagatam Das, Indian Statistical Institute, Kolkata. Email: swagatam.das@isical.ac.in

Prof. P B Sujit, IISER Bhopal. Email: sujit@iiserb.ac.in

Prof. Matthew Taylor, University of Alberta. Email: matthew.e.taylor@ualberta.ca